

Assignment 6

Mobile App Pentesting: Apk.Allsafe.apk

The screenshot displays the MobSF (Mobile Security Framework) Static Analyzer interface. The browser address bar shows the URL: `mobsf.live/static_analyzer/52a7bf23df56e39a26034304e41108f2/`. The interface is divided into a sidebar on the left and a main content area on the right.

Sidebar (Left):

- Static Analyzer
- Information
- Scan Options
- Signer Certificate
- Permissions
- Android API
- Browsable Activities
- Security Analysis
- Malware Analysis
- Reconnaissance
- Components
- PDF Report
- Print Report
- Start Dynamic Analysis

Main Content Area (Right):

The top navigation bar includes: RECENT SCANS, STATIC ANALYZER, DYNAMIC ANALYZER, API, DONATE, DOCS, ABOUT, and a search bar.

PERMISSIONS Table:

PERMISSION	STATUS	INFO	DESCRIPTION	CODE MAPPINGS
android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all networks.	
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.	Show Files
android.permission.QUERY_ALL_PACKAGES	normal	enables querying any normal app on the device.	Allows query of any normal app on the device, regardless of manifest declarations.	
android.permission.READ_EXTERNAL_STORAGE	dangerous	read external storage contents	Allows an application to read from external storage.	Show Files
android.permission.RECORD_AUDIO	dangerous	record audio	Allows application to access the audio record path.	Show Files
android.permission.WRITE_EXTERNAL_STORAGE	dangerous	read/modify/delete external storage contents	Allows an application to write to external storage.	Show Files
infosecadventures.allsafe.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	unknown	Unknown permission	Unknown permission from android reference	

Showing 1 to 7 of 7 entries

Previous 1 Next

SIGNER CERTIFICATE

Binary is signed

v1 signature: True
v2 signature: True
v3 signature: False
v4 signature: False

X.509 Subject: CN=Android Debug, O=Android, C=US
Signature Algorithm: rsassa_pkcs1v15
Valid From: 2022-06-28 07:49:58+00:00
Valid To: 2052-06-28 07:49:58+00:00
Issuer: CN=Android Debug, O=Android, C=US
Serial Number: 0x1

Hash Algorithm: sha1
md5: ec2ef3ae0793999b0910e5c4a0d40109
sha1: 44a3f4b8ce0e5e6c7daed70949917aa1d4369f7
sha256: b1c87b4083a79ff61f28061df64d2b5cbffe13795a3b59679c3e042cbaadb0ae
sha512: 0d71240e5e3a0527840dca24f65ed5e0f0996a8f1f74723606097bf698d167cb343b8be3800a9b25601bfd8e50437b287b7e87f3766261469571fb40fb2b2b3

PublicKey Algorithm: rsa
Bit size: 2048
Fingerprint: d4dd0cd7153a31571dc15983b541cbc51bd51e2fc7b0191b70b7006a5303250d
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION STATUS INFO DESCRIPTION CODE MAPPINGS

android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all	
---	--------	---------------------	---	--

Dec 9 7:33 AM

interhip_task_Assignment : Static Analysis Mobile Security Framework (| +

mobsf.live/static_analyzer/52a7bf23df56e39a26034304e41108f2/

RECENT SCANSSTATIC ANALYZERDYNAMIC ANALYZERAPIDONATE▼DOCSABOUTSearch

Static Analyzer

Information

Scan Options

Signer Certificate

Permissions

Android API

Browsable Activities

Security Analysis

Malware Analysis

Reconnaissance

Components

PDF Report

Print Report

Start Dynamic Analysis

BROWSABLE ACTIVITIES

Search:

ACTIVITY	INTENT
infosecadventures.allsafe.challenges.DeepLinkTask	Schemes: allsafe://, https://, Hosts: infosecadventures, Path Prefixes: /congrats,

Showing 1 to 1 of 1 entries

Previous1Next

NETWORK SECURITY

HIGH1

WARNING0

INFO0

SECURE0

Search:

NO	SCOPE	SEVERITY	DESCRIPTION
1	infosecadventures.io	high	Domain config is insecurely configured to permit clear text traffic to these domains in scope.

Showing 1 to 1 of 1 entries

Previous1Next

Static Analysis

Mk Mobile Security Framework

mobsf.live/static_analyzer/52a7bf23df56e39a26034304e41108f2/

MobSF

Static Analyzer

Information

Scan Options

Signer Certificate

Permissions

Android API

Browsable Activities

Security Analysis

Malware Analysis

Reconnaissance

Components

PDF Report

Print Report

Start Dynamic Analysis

RECENT SCANS

STATIC ANALYZER

DYNAMIC ANALYZER

API

DONATE

DOCS

ABOUT

Search

CERTIFICATE ANALYSIS

HIGH1

WARNING2

INFO1

Search:

TITLE	SEVERITY	DESCRIPTION
Application signed with debug certificate	high	Application signed with a debug certificate. Production application must not be shipped with a debug certificate.
Application vulnerable to Janus Vulnerability	warning	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.
Certificate algorithm might be vulnerable to hash collision	warning	Application is signed with SHA1withRSA. SHA1 hash algorithm is known to have collision issues. The manifest file indicates SHA256withRSA is in use.
Signed Application	info	Application is signed with a code signing certificate

Showing 1 to 4 of 4 entries

Previous

1

Next

MANIFEST ANALYSIS

HIGH2

WARNING7

INFO0

SUPPRESSED0

Search:

Static Analysis

Mk Mobile Security Framework

mobsf.live/static_analyzer/52a7bf23df56e39a26034304e41108f2/

MobSF

Static Analyzer

Information

Scan Options

Signer Certificate

Permissions

Android API

Browsable Activities

Security Analysis

Malware Analysis

Reconnaissance

Components

PDF Report

Print Report

Start Dynamic Analysis

RECENT SCANS

STATIC ANALYZER

DYNAMIC ANALYZER

API

DONATE

DOCS

ABOUT

Search

NO	ISSUE	SEVERITY	DESCRIPTION	OPTIONS
1	App can be installed on a vulnerable unpatched Android version [android:allowBackup=true]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.	
2	App has a Network Security Configuration [android:networkSecurityConfig=@xml/network_security_config]	info	The Network Security Configuration feature lets apps customize their network security settings in a safe, declarative configuration file without modifying app code. These settings can be configured for specific domains and for a specific app.	
3	Debug Enabled For App [android:debuggable=true]	high	Debugging was enabled on the app which makes it easier for reverse engineers to hook a debugger to it. This allows dumping a stack trace and accessing debugging helper classes.	
4	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.	
5	Activity (infosecadventures.allsafe.ProxyActivity) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.	
6	Activity (infosecadventures.allsafe.challenges.DeepLinkTask) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.	
7	Broadcast Receiver (infosecadventures.allsafe.challenges.NoteReceiver) is not Protected. [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.	